

NORTHEAST IOWA, IA, USA

WMO: 725463

Lat: 43.073N

Lon: 92.611W

Elev: 1125

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 14966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.6	-4.2	-16.9	2.3	-7.0	-11.3	3.1	-3.3	32.1	13.5	28.3	16.6	9.0	310	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	90.1	75.4	86.5	73.8	83.9	72.7	79.2	85.3	77.0	83.6	75.0	81.5	11.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	147.9	83.1	74.9	136.5	80.7	72.7	126.6	79.3	43.5	84.6	41.3	83.2	39.3	81.0	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.5	23.6	20.4	-18.7	93.8	6.2	3.0	-23.1	96.0	-26.8	97.8	-30.3	99.5	-34.8	101.7		
			-17.6	82.0	6.4	2.0	-22.2	83.5	-25.9	84.6	-29.5	85.7	-34.1	87.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	18.7	22.4	33.8	47.9	59.5	69.4	72.2	69.8	63.2	50.3	35.7	23.6
	DBStd	21.36	12.83	12.31	12.41	10.14	8.70	6.26	5.36	5.01	7.91	9.47	10.42	11.65
	HDD50	3824	971	774	526	156	17	0	0	0	5	113	443	819
	HDD65	7135	1436	1193	970	519	215	28	7	13	126	464	880	1284
	CDD50	2850	0	0	24	92	313	581	688	614	402	123	13	0
	CDD65	685	0	0	3	4	45	159	230	162	73	9	0	0
	CDH74	5859	0	0	20	81	472	1385	1970	1199	631	100	1	0
	CDH80	1704	0	0	4	19	145	443	612	287	172	22	0	0
Wind	WSAvg	9.7	10.9	11.0	11.1	11.9	10.9	8.8	7.1	6.3	7.7	9.5	10.5	10.5
Precipitation	PrecAvg	34.2	0.9	0.9	2.0	3.3	4.3	4.9	4.6	4.1	3.7	2.4	1.8	1.1
	PrecMax	51.4	2.0	2.8	5.4	7.7	10.2	11.4	18.5	13.6	12.5	5.8	5.4	3.2
	PrecMin	16.6	0.1	0.0	0.2	0.7	1.6	1.4	0.8	0.7	0.6	0.2	0.0	0.2
	PrecStd	6.9	0.5	0.6	1.1	1.5	1.9	2.5	3.0	3.0	2.7	1.3	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	55.0	75.4	82.0	90.5	92.7	92.6	90.3	90.1	83.7	69.6	55.3
		MCWB	42.5	50.3	62.0	62.8	69.2	75.0	78.8	75.6	72.9	65.2	57.0	51.3
	2%	DB	41.4	47.6	66.0	74.9	83.7	88.8	90.0	86.1	84.5	76.6	62.6	46.0
		MCWB	37.6	44.8	57.2	58.6	65.8	73.0	78.4	76.6	71.3	62.5	53.9	43.0
	5%	DB	37.0	41.4	58.6	69.8	79.3	85.5	86.2	82.4	81.2	71.8	55.2	41.3
		MCWB	35.0	38.1	51.3	55.6	64.0	71.6	76.2	73.9	68.8	60.8	48.4	38.4
	10%	DB	34.4	37.2	50.1	64.2	74.7	81.7	83.5	80.9	77.2	65.9	50.4	36.9
		MCWB	33.1	35.4	45.4	53.0	62.3	69.2	74.8	72.2	66.5	57.9	45.7	35.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	50.9	62.9	65.4	74.6	80.0	83.4	81.0	77.7	71.3	59.9	53.3
		MCDB	48.7	54.5	74.5	78.8	83.2	87.4	89.5	85.9	84.6	78.6	65.7	56.7
	2%	WB	38.5	44.8	58.4	61.3	70.4	76.6	80.1	78.0	74.2	66.5	55.0	43.9
		MCDB	41.4	47.8	64.6	71.5	79.6	84.4	85.5	83.8	80.8	71.7	60.6	45.8
	5%	WB	35.6	38.8	51.5	58.2	67.3	74.1	77.8	75.7	71.8	62.4	50.1	38.8
		MCDB	37.3	41.3	58.2	66.9	75.8	82.0	84.0	81.1	78.8	67.7	55.2	41.5
	10%	WB	33.3	36.0	45.2	54.7	64.6	71.8	75.4	73.6	69.1	58.6	45.9	35.5
		MCDB	34.7	37.9	49.5	62.9	71.3	79.5	82.1	78.7	74.3	65.6	49.8	37.2
Mean Daily Temperature Range	5% DB	MDBR	14.9	15.0	15.3	19.6	19.6	18.8	18.5	18.6	21.5	19.8	16.0	13.5
		MCDBR	17.3	17.0	24.5	29.2	25.8	23.3	20.6	20.4	24.8	26.5	24.0	17.0
		MCWBR	14.7	13.8	15.6	15.9	12.4	11.3	11.9	12.8	13.0	14.6	15.6	13.9
	5% WB	MCDBR	16.4	16.2	23.3	24.4	21.2	19.9	18.4	17.7	20.5	21.2	21.9	16.1
		MCWBR	14.1	13.6	15.8	15.7	12.6	11.9	12.4	12.5	12.4	15.0	15.6	13.6
Clear-Sky Solar Irradiance	taub	0.274	0.299	0.321	0.369	0.397	0.411	0.429	0.410	0.377	0.333	0.294	0.271	
	taud	2.390	2.321	2.392	2.299	2.286	2.287	2.225	2.288	2.346	2.460	2.495	2.428	
	Ebn at Noon	276	285	292	284	278	273	267	268	269	270	267	267	
	Edh at Noon	24	31	33	39	41	41	43	40	35	27	22	22	
All-Sky Solar Radiation	RadAvg	574	904	1200	1469	1741	1933	1986	1701	1381	899	601	446	
	RadStd	59	84	125	140	127	154	102	113	99	120	50	35	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page